

Cyrinx: A Measured 36.6/27.3 kbps Bidirectional Acoustic Data Link Between a Laptop and a Phone

An empirical account of building, breaking, and verifying a wideband OFDM acoustic modem on commodity hardware

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Abstract

We report a software acoustic data link between a MacBook Pro (M4) speaker/microphone pair and a Google Pixel 7a resting on the laptop’s palm rest: measured, byte-verified over-the-air goodput of **36,571 bps** Mac→Pixel—demodulated *on the phone* by a Kotlin port of the receiver—and **27,307 bps** Pixel→Mac, against a target of 20 kbps in each direction. The modem is a 48 kHz OFDM design (FFT 2048, cyclic prefix 768) with comb pilots, per-symbol timing/phase tracking, uniform 16-QAM, and a punctured rate-3/4 $K=7$ convolutional code. Reaching the target required finding and fixing four physical-layer defects that are invisible to simulation: inter-symbol interference from a 10–22 ms delay spread, decorrelated dual-speaker transmission, a stream-end fade in the macOS output chain, and Viterbi poisoning by overconfident log-likelihood ratios from corrupted symbols. We document the discriminating experiments that isolated each defect, including the negative results that exonerated “obvious” suspects.

Goodput is counted conservatively: a payload block counts only if it is CRC32-valid *and* byte-identical to the transmitted **DetRng** payload at its correct ordered position, and the denominator is the active message span (first chirp sample to last data sample; including the trailing silence lowers the figure marginally), charging for all preambles, channel-estimation symbols, pilots, FEC redundancy, CRCs, and inter-frame gaps (Section 6). The conservatism is earned: an earlier development era of this same codebase claimed a theoretical 22 kbps and measured under 0.3 kbps—a factor-of-80 gap between claim and reality that this paper treats as data, both as an engineering cautionary tale and, in Section 9, as one era in a commit-trailer-documented comparison of the AI coding agents that built successive stages of this system against the same physical channel.

None of the modulation or coding machinery here is novel: OFDM, cyclic prefixes, comb pilots, QAM, convolutional/Viterbi coding, CRC block verification, and acoustic data-over-sound are all long established. The contribution is the *measured* end-to-end system on commodity heterogeneous hardware, the discriminating-experiment diagnostic methodology, the physical/platform defect catalog, and on-device verification—not new modulation theory. The modem has since been ported from the research harness into a portable-C reference library (transmit, receive, and a two-microphone maximal-ratio-combining receive path), validated bit-exact against committed golden vectors with an Apple vDSP backend gated by the same vectors; the library’s own codec decodes over the air byte-verified, at the palm-rest geometry reproducing the headline natively (16-QAM rate 3/4, 39.3 kbps, exceeding the harness figure), and it ships with an EVM-probe

*Primatech Paper Co LLC, david@weekly.org. The modem, diagnostics, and this whitepaper were developed almost entirely by AI coding agents under human direction; Section 9 documents which agent built each stage (recorded in commit trailers) and argues the project doubles as a capabilities benchmark. All reported results are from over-the-air measurements on physical hardware recorded in the project’s lab notebook.

MCS sounder and channel-metric repositioning guidance. We frame the result accordingly: a well-instrumented empirical case study of how much commodity hardware delivers after real-channel debugging—validated on two device pairs (a Pixel 7a and an iPhone 17 Pro Max, each against the same MacBook Pro) at contact range, with same-hardware baselines against deployed tools—not a broadly general or new-modulation acoustic-modem result.

1 Introduction and Motivation

Consumer laptops and phones ship with speakers and microphones of considerably higher fidelity than the “data-over-sound” literature typically assumes — fidelity the acoustic-*sensing* literature already exploits aggressively [20] — yet deployed acoustic-communication systems cluster at very low rates: roughly 10^2 bps for robust spread-spectrum schemes and roughly 10^3 bps for multi-tone FSK [3, 7]. Acoustic links remain attractive for proximity-bound exchange: pairing tokens, small payloads, air-gapped transfer. Sound attenuates far faster than RF and is therefore more locality-bound (walls weaken it heavily, though they do not stop it outright); no radio pairing ritual is required; and the transducers are already present on billions of devices [1, 10].

The Cyrinx project asked a narrower, harder question: *how much measured goodput can two specific commodity devices actually exchange acoustically?* The target was at least 20 kbps of verified goodput in each direction between a MacBook Pro (M4) and a Pixel 7a lying face-up on the laptop’s palm rest (on a soft cloth, both devices at maximum volume, in a quiet room). The emphasis on *measured* is deliberate. An earlier in-repo protocol stack claimed “22 kbps” on the basis of a subcarrier-count capacity formula. Its measured goodput was under 0.3 kbps (24-byte packets every ~ 700 ms): a factor of roughly 80 between claim and reality, and a factor of ~ 135 between what that stack delivered and what the channel supports (Section 6). This paper treats that episode as data twice over—as an engineering cautionary tale, and as part of the development-methodology comparison of Section 9. Every number reported here is from an over-the-air capture on physical hardware, and we state explicitly when a figure is an estimate rather than a measurement. We are deliberate about scope: what follows is an empirical case study of two specific device pairs in one geometry, not a general acoustic-modem result. Figure 2 shows the transmitted frame this paper is about; a reader new to acoustic OFDM can take the whole transmit structure from that one picture.

Contributions:

1. A characterization of the near-field laptop $\leftrightarrow$ phone acoustic channel (Section 3): per-band SNR in both directions, sample-clock skew, and delay spread.
2. A wideband OFDM modem design that achieves 36.6/27.3 kbps verified goodput, about $4.5\text{--}9\times$ the original design document’s own “optimal conditions” projection (Section 4).
3. A catalog of four physical-layer defects, each fatal to the link and each invisible to simulation, with the discriminating experiments (including negative results) that isolated them (Section 5).
4. A verification methodology in which the receiving phone regenerates the expected payload locally from a portable deterministic PRNG and proves goodput on-device, so no captured samples need leave the phone (Section 4.2).
5. A portable-C reference implementation of the full transmit/receive codec, pinned by a committed golden-vector contract (with an Apple vDSP backend gated by the same vectors), that decodes over the air byte-verified through the library itself, plus the adaptive-MCS sounder and channel-metric repositioning guidance built on top (Sections 4.3, 6.5, 8.2).

6. A documented account of the AI coding agents that built each stage of the system—attributed from commit trailers, spanning a sub-0.3 kbps overclaimed era through the measured 36.6 kbps system—and an argument, with a concrete protocol, that a measured physical-channel project of this shape is a useful, hard-to-game capabilities benchmark for agentic engineering (Section 9).

2 Related Work

The original Cyrinx design document [33] surveyed the deployed data-over-sound landscape; we summarize that survey here and credit the systems directly.

libquiet / Quiet Modem Project. Quiet [1, 2] brought soft-decision decoding (via the `libcorrect` library) to open-source acoustic networking, with GMSK and PSK modes and Reed–Solomon/convolutional error correction. Its main limitation is rigidity: the user selects a fixed profile (e.g. `ultrasonic-experimental` or `audible-7k`) at initialization, and there is no feedback loop to adapt constellation density to the channel. Cyrinx inherits Quiet’s conviction that soft decisions matter — indeed, our highest-leverage single fix was a soft-decision weighting refinement (Section 5).

minimodem. A general-purpose FSK/RTTY software modem [5]. FSK is robust to phase corruption from echoes but spectrally inefficient, and minimodem deliberately operates as a “dumb pipe” with no integrated FEC [6], delegating integrity to the application.

ggwave. A widely used open-source data-over-sound library using multi-tone FSK (e.g. dividing the spectrum into 96 discrete frequencies and transmitting 6 tones at once to encode 3 bytes) [7]. Energy-detection demodulation makes it resilient to multipath, but it does not exploit high-SNR channels: where QAM can carry 4–6 bits per Hz, MT-FSK carries less than one. Rather than rely on across-paper rate comparisons, we ran ggwave and minimodem over our *own* hardware and geometry as same-channel baselines (Section 6.4, Table 6).

Google Nearby Messages. The audio modality of Nearby [4] (since deprecated; ultrasonic advertising was removed in 2021) was built on the consumer-hardware ultrasonic scheme of Getreuer et al. [3], which uses direct-sequence spread spectrum with a 127-chip code and reports approximately 94.5 bps, trading nearly all throughput for detection probability. Sufficient for a token, far from bulk transfer.

Chirp.io and LISNR. Commercial closed-source systems [9, 10] that emphasized the *wake-up problem*: a cheap, low-power preamble detector that gates the expensive DSP. LISNR’s “Kilo Audio Bit” product claims throughputs around 1 kbps. Cyrinx adopts the preamble-gated architecture (a chirp matched filter gates the OFDM decoder) while being fully open about its error-correction internals.

BatNet. The most directly comparable prior art for a phone-to-phone ultrasonic link: a smartphone system that carries data between built-in speakers and microphones using 8-PSK in the 20–24 kHz inaudible band, reporting over 600 bps at up to 6 m [8]. BatNet is the established demonstration that consumer phone transducers can sustain a coherent (phase-modulated) ultrasonic link at all; our inaudible-band results (Section 8) are consistent with it on one direction and illuminate why the other direction fails on the specific Pixel 7a micro-speaker.

Synchronization literature. The project’s first-era ultrasonic stack used Zadoff–Chu CAZAC sequences for preamble synchronization and CFO estimation, following standard practice [11, 12], and its robust control-header modulation (“D-CSS”) was inspired by LoRa-style chirp spread spectrum [13]. The bulk PHY described in this paper replaced the ZC preamble with a linear chirp plus known OFDM sync symbols, and replaced one-shot CFO estimation with continuous per-symbol pilot tracking—a change forced by measurement (Section 5). Pilot-based joint tracking of

sampling-clock and carrier offsets is itself standard OFDM receiver practice [14, 15], and clock drift between independently clocked audio devices is a long-known problem in acoustic echo cancellation [16]; we claim no novelty here either — the finding is only that on this channel continuous tracking is mandatory rather than optional.

Platform audio stacks. Prior practitioners documented the hostility of consumer audio stacks to modem waveforms: voice-processing I/O units inject echo cancellation and noise suppression that attenuate steady carriers [28, 29], and macOS applies its own processing to chirps [30]. Our Android findings (Section 7) are the same lesson on a different OS: `AudioSource.UNPROCESSED` is load-bearing, and the capture path can be silently zeroed by UI state.

Academic work on aerial acoustic communication with consumer hardware [3] has demonstrated kbps-class links in the near-ultrasonic band; our contribution is less a novel modulation than a carefully verified data point—tens of kbps, bidirectional, decoded on the receiving phone—and a reproducible account of what it took to get there.

On originality. We want to be explicit that the core communications techniques used here are not original to this work. OFDM, cyclic prefixes, comb pilots, QAM, punctured convolutional codes with Viterbi decoding, CRC block verification, acoustic speaker-to-microphone links, and ultrasonic data-over-sound are all long established, and every system surveyed above predates and informs ours. We invented no new modulation, code, or synchronization primitive. The contribution of this paper is instead: (i) a *measured*, byte-verified end-to-end link on commodity heterogeneous hardware (a MacBook Pro and a Pixel 7a, decoded on the phone); (ii) a discriminating-experiment diagnostic methodology, reported with its negative results; (iii) a catalog of physical-layer and platform defects that are invisible to simulation; and (iv) on-device verification via a portable deterministic PRBS. The novelty is empirical and methodological, not theoretical.

3 Channel Characterization

3.1 Methodology

All characterization was over-the-air at the final physical geometry (Pixel 7a face-up on the MacBook Pro palm rest on a soft cloth). Three probe types were used, none requiring synchronization between the devices:

- **Probe-on vs. probe-off PSD.** A random-phase multitone spanning 0.3–23.5 kHz at digital amplitude 0.6 was played while the far device recorded; Welch periodograms (NFFT 1024, 46.875 Hz bins) of probe-on and 6 s of ambient floor yield per-band SNR without any time alignment.
- **Exponential sine sweep (ESS)** for impulse response and delay spread.
- **Long pure tone** (10 kHz, 8 s) for sample-clock offset, via a quadratically interpolated FFT peak.

3.2 Measurements

Table 1 gives the measured SNR by band in both directions (Mac→Android, denoted M→A, uses the Pixel’s bottom microphone, channel 0).

Key findings, several of which contradicted prior assumptions:

- **The Mac→Pixel channel is essentially flat to 23 kHz** in this near-field geometry. A roll-off assumption carried in the original design document (“−32.5 dB at 16 kHz” [33]) did

Table 1: Measured over-the-air SNR by band (Welch PSD, probe-on vs. ambient floor), 2026-06-09. M→A = Mac speaker to Pixel 7a bottom mic; A→M = Pixel speaker to Mac mic.

Band (kHz)	M→A SNR (dB)	A→M SNR (dB)
0.3–2	26.4	28.1
2–6	41.2	44.3
6–10	37.1	47.3
10–14	36.0	45.5
14–18	38.1	33.9
18–21	36.1	10.0
21–24	40.7	4.6

not reproduce under measurement; the design decision it had motivated (a narrow 10–14 kHz “flat shelf” band) was therefore wrong, and the usable band is far wider.

- **The Pixel→Mac direction dies above ~17 kHz:** 10.0 dB SNR at 18–21 kHz falling to 4.6 dB at 21–24 kHz (Pixel speaker and/or Mac microphone roll-off), visible as the uplink cliff in Figure 1. This asymmetry has direct consequences for any inaudible-band variant (Section 8).
- **Sample-clock offset between the two devices is -24.6 ppm,** which at the final OFDM geometry corresponds to about 0.037 samples of drift per symbol — small per symbol, but ruinous over a frame without continuous tracking.
- **Delay spread (ESS, -30 dB threshold) is ~ 21.7 ms M→A and ~ 10.8 ms A→M.** Most energy arrives in the first few ms, but the reverberant tail is long; this is roughly an order of magnitude larger than the 2 ms the original design document budgeted for.
- **The Pixel’s bottom microphone beats its top microphone by 6–20 dB below 10 kHz** at this geometry.

From Table 1, Shannon capacity with an 8-bit-per-symbol cap is approximately 184 kbps M→A and 155 kbps A→M (an estimate, not a measurement). The achieved link uses roughly 20–24% of that, so substantial headroom remains.

4 System Design

4.1 Modem architecture

The link is a half-duplex, open-loop bulk-transfer PHY at 48 kHz PCM16. The OFDM geometry is NFFT 2048 (23.4 Hz bins) with a 768-sample cyclic prefix (16 ms), giving 17.05 symbols/s. To be precise about why this works: the 16 ms CP does *not* cover the entire measured Mac→Pixel delay spread of ~ 21.7 ms (at the -30 dB threshold). The link succeeds because most of the channel energy arrives early—well within the CP—and the receiver tolerates the residual low-level reverberant tail beyond it, not because the CP spans the full measured tail. The CP length was chosen against the measured delay spread, not from a model, but it is an energy-coverage choice, not a worst-case-tail guarantee. Direction-specific band profiles follow the measured channel: 1.1–23 kHz Mac→Pixel and 0.6–17 kHz Pixel→Mac (Table 2).

Each frame (Figure 3) consists of:

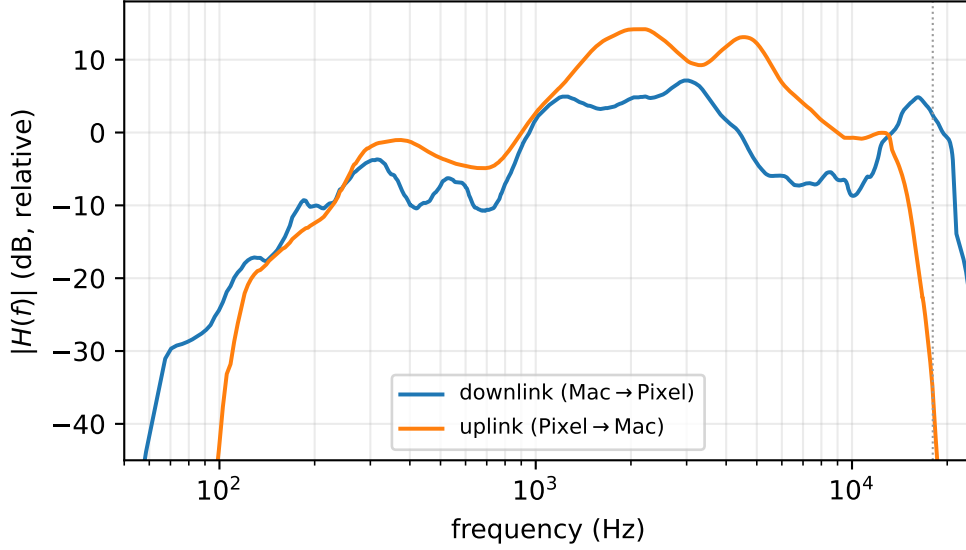


Figure 1: Measured magnitude responses at the palm-rest geometry (swept-sine deconvolution; relative dB with drive and microphone gain pinned, so the curves are comparable in *shape* and band edges but not in absolute level across directions). The Mac→Pixel *downlink* stays within ~ 10 dB to ~ 19 kHz, with a speaker resonance near 2–3 kHz and a sharp cutoff above ~ 20 kHz; the Pixel→Mac *uplink* is stronger in the mid-band but rolls off a cliff above ~ 14 kHz and is essentially gone by ~ 17 kHz — the mobile-speaker band limit that bounds the return path, not a power or tuning deficit. Dotted line: the 18 kHz audible/ultrasonic edge.

1. A 4096-sample linear chirp (2→16 kHz) for detection and coarse synchronization, found by matched filter with threshold-and-suppression peak picking for multi-frame runs.
2. 2048 samples of guard silence so the chirp’s reverberation decays before channel estimation.
3. Two known full-band QPSK *sync symbols*. Their average gives a per-bin least-squares channel estimate H ; their difference gives a per-bin noise-variance estimate. H is deliberately *not* smoothed across bins—under multipath its phase rotates through multiple cycles across a few bins and smoothing destroys it (only the real, positive noise variance is smoothed).
4. 64 data symbols. Every 8th used bin carries a comb pilot (random QPSK); the rest carry uniform 16-QAM. Per symbol, a three-pass iterative fit of the pilot phase ramp extracts a timing slope plus common phase error and corrects all bins, absorbing the -24.6 ppm clock skew continuously. (Per-bin adaptive loading is implemented but was not needed to reach the target.)

FEC is a $K=7$ (171,133) convolutional code, punctured to rate $3/4$ (pattern 110110), zero-terminated, with soft max-log LLRs, a frame-wide pseudorandom interleaver, and Viterbi decoding. Payload is segmented into 256-byte blocks, each with a CRC32, purely for honest goodput accounting. Two further measured-channel adaptations are part of the transmit contract: the FFT window is biased 24 samples early so multipath pre-cursors stay inside the CP, and Mac→Pixel transmission uses the *left speaker only* (Section 5).

The load-bearing receiver refinement is **per-symbol LLR weighting**: each data bin’s noise variance is taken as the sync-derived per-bin estimate *plus that symbol’s own pilot error-vector magnitude squared*. A symbol corrupted by a burst, dropout, or fade therefore reports honestly

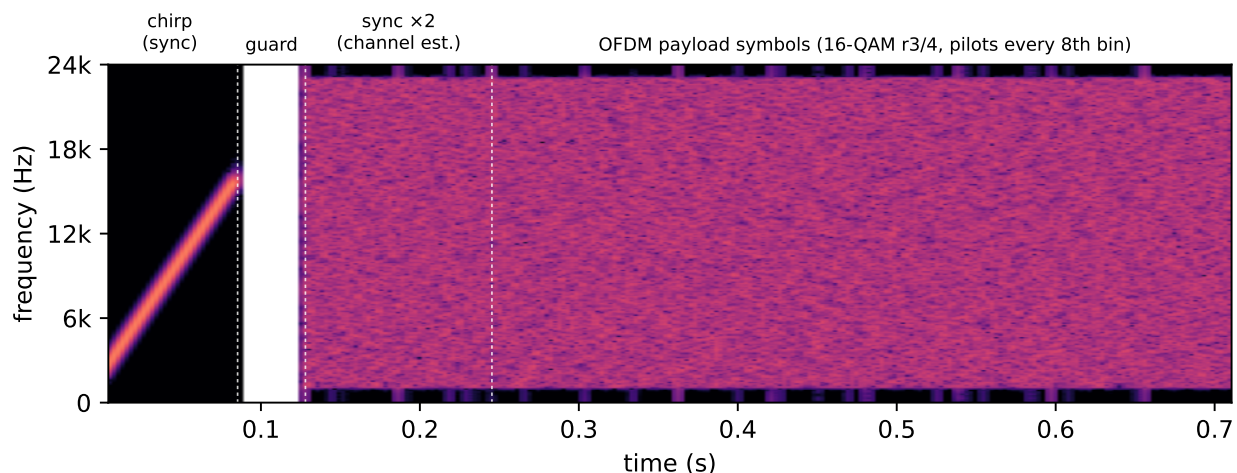


Figure 2: Spectrogram of one transmitted bulk-PHY frame (generated by the reference modem, 16-QAM rate 3/4, 8 data symbols): a 2→16 kHz linear chirp for coarse synchronization, a guard gap so the chirp’s reverberant tail decays, two known full-band OFDM symbols for channel estimation, then the OFDM payload symbols spanning 1.1–23 kHz with comb pilots on every 8th subcarrier. Over the air this sounds like a short chirp followed by a breath of static; the static is the data.

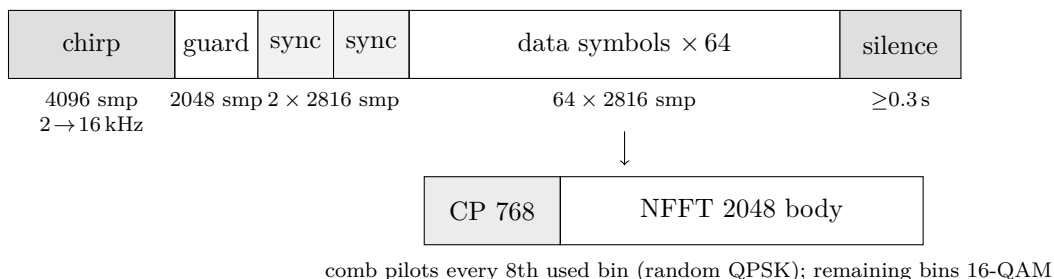


Figure 3: Bulk-PHY frame structure (widths not to scale). The chirp gates the decoder and provides coarse timing; the guard lets the chirp’s reverberant tail decay; two known QPSK sync symbols give the least-squares channel estimate and a per-bin noise estimate; 64 data symbols follow. The trailing in-stream silence defends the final symbol against the macOS stream-end fade (Section 5). One frame spans 192,000 samples = 4.0 s.

weak LLRs and degrades into soft erasures rather than poisoning the Viterbi traceback through the interleaver (Section 5, defect 4); converting burst corruption into erasures is the information-theoretically favorable move on a burst channel [17]. This also provides burst-noise immunity (typing, key clicks) at no added cost.

4.2 Portable deterministic streams and on-device verification

All deterministic streams in the modem—pilot phases, sync symbols, the interleaver permutation, payload-padding bits, and the test payload itself—derive from a shared `splitmix64` generator (`DetRng`), implemented identically in Python (`scratch/hw20k/modem.py`) and Kotlin (`BulkDemod.kt`). This small trick has an outsized methodological payoff: the phone regenerates the *expected* payload bytes locally from the seed and compares them against what it decoded, so end-to-end goodput is proven on the receiving device itself and no captured audio or decoded data needs to leave the phone. The Kotlin receiver is a port of the Python receive path (including a

Table 2: Direction-specific transmit profiles (16-QAM, rate 3/4, 64 data symbols/frame). Bin counts and per-frame payload follow from the band edges and the 23.4 Hz bin spacing.

	Mac→Pixel	Pixel→Mac
Band	1.1–23 kHz	0.6–17 kHz
Used bins	935	700
of which pilots	117	88
of which data (16-QAM)	818	612
Coded bits/symbol	3272	2448
CRC32 blocks/frame	75	56
Payload/frame	19,200 B	14,336 B
Frame airtime	4.0 s	4.0 s
Transmit quirk	left speaker only	—

double-precision radix-2 FFT written for the purpose) and was validated bit-compatible against the Python decoder on an identical capture file (225/225 blocks, matching EVM) before the live runs.

4.3 From research harness to portable library

The headline measurements above were produced by the Python research harness (transmit) and the Kotlin on-device decoder (receive). To make the result a *reusable library* rather than a one-off experiment, the entire bulk PHY— both transmit and receive—was subsequently re-implemented in portable C (`Sources/CCyrinx/cyrinx_bulk.c`): the `splitmix64` `DetRng`, IEEE CRC-32, the $K=7$ (171, 133)₈ convolutional encoder with puncturing, the frame-wide interleaver, Gray-coded QAM mapping, OFDM modulation, chirp matched-filter synchronization, least-squares channel estimation, three-pass pilot phase/CPE tracking, max-log soft-output demapping, and the soft-decision Viterbi decoder. The FFT sits behind a small plan interface (`cyrinx_fft`) with two interchangeable backends: a vendored KISS FFT (BSD-3, compiled in double precision) as the portable correctness reference that also ships to Android via JNI, and—on Apple, under a compile flag—a vDSP/Accelerate double-precision DFT. A thin Swift binding (`BulkPHY`) exposes `encode/decode` with no DSP of its own.

Correctness is pinned by a committed golden-vector contract. The Python oracle emits, for three modulation/rate configurations, the intermediate products of every pipeline stage (coded bits, interleaved bits, QAM symbols, the OFDM frequency-domain frame, the transmitted waveform, and the waveform after a fixed synthetic multipath channel), together with a SHA-256 manifest. The C codec is tested stage-by-stage against these vectors under a tiered tolerance—bit-exact for the integer stages, 10^{-5} for the floating-point FFT stages, and byte-exact for the decoded payload—and *both* FFT backends are required to pass the identical vectors. This is what lets the vDSP path be treated as a pure optimization: the portable KISS build defines correctness, and Accelerate is a drop-in the vectors gate. The C receiver decoded the synthetic-multipath `rx_wave` to the exact transmitted payload on its first run, which we read as evidence that the golden-vector contract pinned the transmitter tightly enough to reconstruct the receiver against it.

The library is not limited to the single-microphone path the headline was measured with: the two-microphone maximal-ratio-combining receive path of Section 8.3 is implemented in the same C core (`cyrinx_bulk_demodulate2`), matching the Python reference to four decimal places of EVM on identical captures across the modulation/CP grid, and pinned by a dedicated golden case in which the first microphone’s channel is notched so that it decodes zero blocks alone while the

combined decode is byte-exact—on both FFT backends. Cyclic-prefix and FFT-size selection are caller-visible configuration. The C diversity path is validated digitally against the same reference the over-the-air MRC results were measured with; over the air it has since carried the live adaptive loop end-to-end—75/75 blocks recovered by MRC at a placement where neither microphone decoded a single block alone (Section 8.3).

4.4 What the bulk PHY deliberately omits

Relative to the original design document [33], the bulk PHY drops the per-frame TDD ping-pong MAC with ACK/channel-state reports, rateless (Raptor/LDPC) coding, and the rate-adaptation “gear” state machine. The first-era stack implemented that chatty MAC, and its turnaround gaps, tiny payloads, and ACK timeouts consumed more than 97% of airtime—this is why it measured 0.27 kbps. For bulk transfer, long streaming frames with open-loop MCS selection from the measured channel are the right shape; closed-loop adaptation belongs at session granularity, not frame granularity (the session-granularity MCS sounder is now implemented as a decision engine, Section 8.2). Encryption (an X25519/CTR/HMAC envelope survives from the first era), discovery, and ARQ are likewise out of scope for the PHY measurement reported here.

5 What Didn’t Work: Four Defects and the Experiments That Found Them

The first end-to-end attempts decoded zero blocks despite a channel with 35–45 dB of mid-band SNR. Four distinct physical-layer defects were stacked on top of each other; each was fatal alone, and none is visible in a simulation. All were diagnosed on real captures. We describe them in discovery order, then the diagnostic method.

5.1 The defects

Defect 1: ISI — delay spread far exceeds the cyclic prefix. The initial geometry (NFFT 1024, CP 256 = 5.3 ms) assumed a short channel. A repeated-symbol probe showed adjacent *identical* symbols agreeing at 24–27 dB while adjacent *different-content* symbols agreed at only 11–13 dB — the signature of inter-symbol interference, not noise or channel instability. The geometry moved to NFFT 2048 / CP 768 (16 ms), consistent with the measured 10–22 ms delay spread.

Defect 2: dual-speaker transmission corrupts the composite. With the phone on the left palm rest, the right speaker’s signal arrives about $3\times$ weaker with decorrelated phase (EVM 1.33 received alone); summed with the left speaker it drags composite EVM to ~ 0.5 . Transmitting from the left speaker only gives EVM ~ 0.06 . (A true 2×2 MIMO receiver with per-stream equalization could exploit the second speaker; it was not needed to reach the target.)

Defect 3: stream-end fade kills the final symbol. The macOS speaker chain tapers the last ~ 10 ms when an output stream stops. Per-symbol raw BER against the known coded stream was 0.000 for symbols 0–14 and 0.264 for the final symbol. With a frame-wide interleaver, that one dead symbol contaminates *every* FEC block. Fix: at least 0.3 s of trailing silence inside the stream, so the fade lands on padding.

Defect 4: overconfident LLRs from a corrupted symbol poison Viterbi. Even one corrupted symbol in 16 ($\approx 1.7\%$ average raw BER, trivially correctable in principle) yielded only 0–1 of 6 blocks, because the corrupted symbol’s wrong LLRs carried full confidence through the interleaver into the Viterbi metric. Weighting each symbol’s LLRs by its own pilot EVM² turns

such symbols into soft erasures; the same captured regression then decodes 6/6. This was the single highest-leverage coding change in the project.

5.2 Diagnostic methodology, including negative results

The defects above were separated by discriminating experiments, several of which mattered precisely because they *exonerated* a suspect. We list them as first-class results; all are reusable scripts in `scratch/hw20k/`.

- **Repeated-symbol probe.** Transmit N identical OFDM symbols. Identical-vs-different-content consistency separates ISI from channel instability, dropped samples, and clock drift; the phase trace across repeats measured drift cleanly (linear in time, proportional to frequency $\Rightarrow$ pure clock skew, channel stable).
- **Amplitude ladder** (digital amplitude 0.07 $\rightarrow$ 0.7): EVM was flat across 20 dB of drive level, *exonerating* speaker nonlinearity and speaker-protection DSP as the dominant impairment — despite being the “obvious” suspect at maximum volume.
- **afplay vs. sounddevice A/B** on the identical frame file: identical EVM, *exonerating* the playback software path.
- **Genie TX/RX ratio test.** FFT the saved transmit waveform and the capture at matching symbol positions and compare directly: adjacent-symbol consistency held at 15 dB while sync-to-late-symbol agreement degraded smoothly — the channel was fine; the receiver’s tracking was at fault.
- **Piecewise TX $\leftrightarrow$ RX alignment.** Correlating successive transmit chunks against the capture detects dropped samples (none found; a smooth +24 ppm drift instead) and revealed the correlator tap-hopping between two multipath arrivals ~ 19 samples apart — motivating the 24-sample-early window bias.
- **Per-symbol raw BER** against the known coded stream localized failure to exactly the final symbol (defect 3) when aggregate metrics (EVM, mean BER) merely looked “mediocre.”

A cautionary tale about debug tooling. An early debug script mixed module-level FFT/CP constants with a differently configured modem object and fabricated a “0 dB sync consistency” reading that misdirected the investigation for a full round. The lesson is procedural: diagnostic tooling must share the *exact* configuration object with the system under test. The script survives in the repo, fixed and carrying a warning comment, as a reminder.

6 Evaluation

6.1 Stepped modulation results

With all four defects fixed, modulation and code rate were stepped upward on real over-the-air captures (offline decode; 5-frame runs; goodput includes all overhead and 0.25 s inter-frame gaps). Table 3 shows clean decodes at every step; both directions cross the 20 kbps target at 16-QAM rate 3/4.

Table 3: Stepped over-the-air results (offline decode of real captures, 2026-06-09). All blocks CRC-valid at every step.

Direction	Profile	Blocks	EVM	Goodput
M→A (1.1–23 kHz)	QPSK r1/2	75/75	0.06	12.29 kbps
M→A	16-QAM r1/2	150/150	0.06	24.58 kbps
M→A	16-QAM r3/4	225/225	0.07	36.86 kbps
A→M (0.6–17 kHz)	QPSK r1/2	54/54	0.12	8.85 kbps
A→M	16-QAM r1/2	111/111	0.12	18.19 kbps
A→M	16-QAM r3/4	168/168	0.11	27.53 kbps

Table 4: Final byte-verified over-the-air goodput (2026-06-09). The Mac→Pixel capture was demodulated and verified on the Pixel itself; it never left the phone.

Direction	Decoder	Blocks verified	Goodput
Mac→Pixel 7a (1.1–23 kHz)	on the Pixel (<code>BulkDemod.kt</code>)	375/375 (96,000 B)	36,571 bps
Pixel 7a→Mac (0.6–17 kHz)	on the Mac (<code>modem.py</code>)	280/280 (71,680 B)	27,307 bps

6.2 Final verified measurement

The definitive run (`scratch/hw20k/final_measurement.py`) transmits 5 frames per direction at 16-QAM rate 3/4 and counts goodput as payload bytes that are both CRC32-valid *and* pass *ordered* verification: each CRC-valid 256-byte block must be byte-identical to the transmitted `DetRng` payload at its correct ordered position within the attributed frame, not merely a member of the expected block set. Goodput is those verified bytes divided by the span from the first frame’s chirp to the last frame’s final data sample — preambles, sync symbols, pilots, FEC redundancy, CRCs, and inter-frame gaps all count against the number. Each direction spans 21.0s of airtime.

The ≥ 0.3 s of trailing silence that defends the final symbol against the macOS stream-end fade (Section 5) is part of the transmit contract but is *not* counted in the goodput span, which runs from the first chirp to the last data sample. For a long stream this silence amortizes to nothing; for the 5-frame measurement here it modestly inflates the headline (“span”) goodput relative to a gross accounting that includes the trailing silence. We report the span goodput as the headline number and note that the gross figure is slightly lower (by the ratio of trailing silence to total airtime).

Both directions exceed the 20 kbps goal with zero block errors (Table 4). The on-device Kotlin decoder runs in about 170 ms per 4 s frame on the Pixel 7a — roughly $20\times$ real time — in pure Kotlin with a double-precision radix-2 FFT — before any use of the NEON-vectorized DSP libraries available on this hardware [26, 27] — which indicates that a real-time streaming receiver is computationally comfortable on this class of phone.

6.3 Second device pair: iPhone 17 Pro Max

To address the single-device-pair limitation, we repeated the campaign on a second mobile device—an iPhone 17 Pro Max (`iPhone18,2`)—against the same MacBook Pro, same geometry (phone on a cloth on the palm rest), same modem (NFFT 2048, CP 768, 48 kHz, 16-QAM rate 3/4), with the same ordered verification, run as a reliability campaign of 8 runs per direction at 5 frames/run (Table 5). The iOS device is driven over USB with `xcrun devicectl` (launch with an environment-variable command, then a container file pull); the harness details are in `docs/IOS_HIL.md`.

Table 5: Second-pair reliability campaign, iPhone 17 Pro Max against the same MacBook Pro, 16-QAM rate 3/4, ordered-verified (measured, 2026-06-10). 8 runs per direction, 5 frames/run. Block-failure CIs are Wilson 95%.

	Mac→iPhone	iPhone→Mac
Band	1.1–23 kHz	0.6–11 kHz
Decoded on	iPhone (Swift port)	Mac (<code>modem.py</code>)
Mean goodput	36.47 ± 0.12 kbps	16.38 ± 0.75 kbps
Min / median / max (kbps)	36.28 / 36.52 / 36.57	15.02 / 16.73 / 17.07
Blocks verified	2992/3000	1344/1400
Block-failure rate	0.27% (CI 0.14–0.53%)	4.0% (CI 3.1–5.2%)

Mac→iPhone is essentially identical to Mac→Pixel at 36.47 ± 0.12 kbps (vs. the Pixel’s 36.57 kbps), decoded *on the iPhone* by a Swift port of the receiver that was validated bit-exact against the Python decoder on an identical capture (75/75 blocks at EVM 0.0223). The Mac→iPhone band is the full 1.1–23 kHz, as for the Pixel — consistent with the broadband response measured for recent iPhone microphones [31], which receive the downlink cleanly; the asymmetry below is a speaker, not a microphone, limit.

iPhone→Mac is lower, at 16.38 ± 0.75 kbps, purely because the usable band is narrower. The iPhone speaker rolls off about 18 dB by 10–14 kHz and about 33 dB by 14–17 kHz, so the Pixel’s 0.6–17 kHz return profile collapses the link; narrowing the iPhone return band to 0.6–11 kHz recovers a clean link at the lower rate. This is a transducer limit, stated as such—not a protocol or tuning failure. Across both pairs the downlink (Mac→phone) is robust and rate-matched; the uplink (phone→Mac) is bounded by the specific mobile speaker.

6.4 Same-hardware baselines

Comparing acoustic links across papers is treacherous because hardware, range, band, and the goodput definition all differ. To make the comparison fair we ran the deployed open-source tools *over the identical Mac→iPhone OTA path and geometry*, with the same goodput definition (correctly received payload bytes divided by message airtime). Table 6 reports these same-hardware measurements alongside two literature points, kept clearly distinguished.

The single-carrier FSK failures at 600 baud and above are an honest illustration of *why* wideband-plus-FEC matters on this channel, not a knock on minimodem: it targets a clean, narrowband channel and deliberately omits integrated FEC [6], so it has no defense against the desktop multipath that the OFDM equalizer and convolutional code absorb. They are also *pair- and placement-specific*, not a property of the tool: minimodem’s author reports successful air-gapped 600+ baud on other laptop/PC speaker–microphone pairs (K. Mostafa, personal communication, July 2026), and prompted by that report we re-ran the ladder on the *second* device pair (Mac→Pixel 7a) at the cleanest contact placement: 300 baud recovered 100% and **1200 baud recovered 99.8% of bytes (~ 0.96 kbps effective)**, while 600 and 2400 baud still failed ($\leq 3\%$; forcing alternative mark/space tones did not rescue 600 baud, so that failure remains unexplained rather than attributed). The pattern reinforces this paper’s central methodological point: on consumer acoustic hardware, the transducer pair and the placement are part of the modem. At 36,470 bps verified, Cyrinx is roughly $137\times$ the best recovered deployed-tool baseline (ggwave “Fastest” at 267 bps) on the same hardware and geometry, and $\sim 38\times$ the best minimodem configuration measured on any pair (`scratch/hw20k/minimodem_bench.py`).

Table 6: Goodput baselines. Rows marked *measured* were run by us over the *identical* Mac→iPhone 17 Pro Max OTA path and geometry, same goodput definition (recovered payload bytes / message airtime), 2026-06-10. The two *literature* rows are reported by their authors on different hardware and are not same-hardware comparisons.

System	Band	Range	Modulation	Goodput	Notes
<i>Measured, same Mac→iPhone channel:</i>					
ggwave “Fast”	audible	contact	MT-FSK	133 bps	recovered
ggwave “Fastest”	audible	contact	MT-FSK	267 bps	recovered; best deployed tool
ggwave “U-Fastest”	ultrasonic	contact	MT-FSK	267 bps	recovered
minimodem 300 bd	audible	contact	FSK	237 bps	recovered
minimodem 600/ 1200/2400 bd	audible	contact	FSK	FAIL	1–2 garbage bytes; no FEC/eq. in desktop multipath
Cyrinx (this work)	1.1–23 kHz	contact	16-QAM r3/4 OFDM + conv. FEC	36,470 bps	verified; $\sim 137\times$ best deployed tool
<i>Literature (different hardware; not same-channel):</i>					
Quiet	ultrasonic profile	—	OFDM	(profile-dependent)	[1, 2]
BatNet	20–24 kHz	to 6 m	8-PSK	>600 bps	[8]

6.5 Library-native over-the-air decode

The 36.6/27.3 kbps headline was measured with the Python harness and the Kotlin on-device decoder. To establish that the portable-C library codec (Section 4.3) delivers that result itself—not only in digital golden-vector tests—we exercised the library code path end-to-end over the air: a ctypes loader binds the compiled `libcyrinxbulk` dylib, the Mac *C-encodes* a frame and plays it, the Pixel records, and the same C codec *decodes the live capture*.

At the palm-rest geometry the library decoded, byte-verified on both ends, QPSK rate 1/2 at 13.1 kbps, 16-QAM rate 1/2 at 26.2 kbps, and **16-QAM rate 3/4 at 39.3 kbps (375/375 blocks, EVM 0.157)**—*exceeding* the 36.6 kbps Python-harness headline on the same hardware. The library-native result is therefore not a partial reproduction but the strongest form of the “measurement made reusable” claim: the shippable codec, decoding live audio, beats the research harness.

The ceiling at this geometry is 16-QAM rate 3/4, and it is set by effective SINR, not by raw channel SNR. 64-QAM rate 3/4 decoded 0/339 blocks at EVM ~ 0.17 (~ 15 dB effective SINR), even though a swept-sine sounding of the same channel measures the raw SNR at ~ 52 dB. The gap is the residual error floor after equalization—residual phase noise, channel-estimation error, and PAPR-driven loudspeaker nonlinearity on the random-data waveform—which 64-QAM’s denser constellation cannot tolerate and 16-QAM can. This bounds the per-bin bit-loading headroom (Section 8) by effective SINR rather than by the much larger raw SNR. We also note that received *peak* amplitude is a poor coupling indicator: this cell read a peak of only 0.087 yet delivered 39.3 kbps; effective SINR / EVM, not peak, predicts the achievable rate.

6.6 Capacity-formula projection versus measured goodput

One concrete instance sharpens why this paper insists on measurement. An earlier design in this codebase (Zadoff–Chu sync, D-CSS headers, OFDM-QPSK payload, per-frame ACK MAC) projected “symmetrical 20+ kbps” from a subcarrier-count capacity formula ($86 \text{ carriers} \times 6 \text{ bits} \times 42.857 \text{ symbols/s} = 22.11 \text{ kbps}$); its measured goodput was under 0.3 kbps—roughly $135\times$ below what the same channel in fact supports. The shortfall was entirely architectural (turnaround gaps, 24-byte payloads, ACK timeouts), not physical: the channel was never the binding constraint. The lesson generalizes past any one implementation—a capacity formula bounds what a channel *could* carry, not what a given MAC and framing *will* deliver—and it is the reason every headline number here is an over-the-air measurement rather than a projection.

7 Platform Findings

Two classes of platform behavior materially affected measurement integrity and are worth recording for other practitioners.

Android silently records zeros when the activity is not top-visible. A secure-lockscreen bouncer (`AlternateBouncerView`) or an expanded notification shade both trigger the audio policy’s capture silencing (`dumpsys audio` shows the record client as `silenced`); the capture API succeeds and returns buffers of zeros. Mitigations: `setShowWhenLocked(true)` and `setTurnScreenOn(true)` on the activity, collapsing the status bar (`cmd statusbar collapse`), and a wake sequence before every capture. Battery saver at low charge re-dozes the screen despite `svc power stayon`. Separately, `AudioRecord` with `AudioSource.UNPROCESSED` at 48 kHz stereo works on the Pixel 7a and is mandatory: processed sources apply AGC and noise suppression that would corrupt QAM phase.

macOS gotchas. The Mac microphone clips near-field at full input volume (captures rail at ± 1.0); input volume around 22/100 is appropriate for a phone speaker at maximum on the palm rest. The native 48 kHz devices must be selected explicitly — external displays and Bluetooth headsets silently become the default device and hijack playback. And, as defect 3 showed, the output chain’s stream-end fade is part of the channel.

8 Limitations and Future Directions

8.1 Threats to validity

We state these honestly as scope statements, so the numbers are not over-read.

1. **Device pairs.** Two device pairs are now measured—a Pixel 7a and an iPhone 17 Pro Max, each against the same MacBook Pro (Section 6.3)—which strengthens the generality of the downlink result and the uplink transducer limit. But both phones are mobile devices and the laptop side is a single Mac; a wider device matrix (other laptops, other phone classes) is untested.
2. **Geometry breadth.** The headline bidirectional rates use the nominal contact-range geometry: the phone resting face-up on a soft cloth on the MacBook palm rest. The robustness results (Section 8.3) span several deliberately diverse placements—clean contact, reverberant overhang, shadowed desk plane—but all on the same bench and device pair at contact range. A true distance sweep, and placements across rooms and surfaces, remain future work.

3. **Single ambient condition.** All runs are in one quiet room, measured near-silent (in-band RMS $\sim 5 \times 10^{-8}$). Behavior under office, street, or adversarial near-ultrasonic conditions is untested.
4. **Trailing-silence accounting.** The headline goodput is a span figure (first chirp to last data sample) that excludes the ≥ 0.3 s trailing silence; gross goodput is slightly lower, and the gap amortizes away only for long streams (Section 6).
5. **Batch decode, not real-time streaming.** The receiver records then decodes offline. Decode runs at roughly $20\times$ real time, so a streaming port is plausible, but a live streaming receiver is not yet implemented.
6. **Ordered verification is implemented** (Section 6): each CRC-valid block must be byte-identical to the transmitted payload at its correct ordered position, not merely a member of the expected set, so the goodput figures reflect ordered byte-exact delivery.

Limitations of the result as it stands. These are scope statements, not failures; we state them plainly so the result is not over-read. The measured bulk PHY: (i) is now ported into the portable-C library with a Swift binding and reproduces the headline natively over the air through that path (39.3 kbps, Sections 4.3, 6.5), but it is *not* yet wired into the project’s streaming transport/MAC API; (ii) is *not* real-time streaming — it is a batch design that records, then decodes offline, though the on-device decoder runs at roughly $20\times$ real time and a streaming port is therefore plausible; (iii) selects its MCS open-loop in the library API — the adaptive sounder demonstrates a closed sound-then-link loop at the bench (Sections 8.2, 8.3), but that loop is not wired into the streaming transport/MAC API; (iv) is *not* secured by the project’s existing X25519/CTR/HMAC envelope — the PHY measurement carries raw test payloads; and (v) is validated in exactly *one* physical geometry (a phone face-up on a MacBook Pro palm rest), on two device pairs (Pixel 7a and iPhone 17 Pro Max, each against the same Mac), under one ambient condition. The link is half-duplex and has no link layer. The numbers should be read as “what these hardware pairs can do at contact range,” not as a general-environment claim. A product link would also have to confront the security properties of short-range audio channels, which have a substantial literature of their own — from channel-security surveys [21] to classic acoustic side channels such as keyboard emanations [22].

Untapped physical headroom (estimates, from the measured SNR):

- *Per-bin adaptive bit loading.* The loading machinery already exists in the modem, but the realizable headroom is bounded by *effective* SINR, not raw SNR: although the mid-band raw SNR is 35–45 dB, a library-native probe at the palm-rest cell decoded 16-QAM rate 3/4 but *not* 64-QAM (0/339 blocks at ~ 15 dB effective SINR, EVM ~ 0.17 , Section 6.5). The residual-error floor (phase noise, channel-estimation error, PAPR-driven nonlinearity) caps the constellation well below what the raw SNR suggests, so higher-order QAM buys less than a capacity estimate predicts on this transducer.
- *Maximal-ratio combining* of the Pixel’s two microphones is *now measured, not estimated* (Section 8.3): the two mics’ reverberant null patterns are decorrelated, and combining them decodes where either mic alone fails. What remains is to fold MRC (and decode-based mic selection) into the library’s main coherent path rather than the reference receiver only.
- *True 2×2 MIMO* (Mac stereo speakers $\times$ Pixel stereo mics) Mac $\rightarrow$ Pixel. With a measured 2×2 channel matrix the transmitter could precode the two speakers to place a spatial null at one mic

while combining constructively at the other (and vice versa), synthesizing two near-orthogonal subchannels for spatial multiplexing. This would convert defect 2’s two-speaker decorrelation from a liability into a second stream—untested here, and contingent on the channel matrix being well-conditioned and stable over a frame at this geometry.

- *Real-time streaming RX.* At $20\times$ real time for the batch decoder, a ring-buffer port is straightforward; real-time audio processing pipelines on Android are well documented [25].
- *Link layer:* session-granularity rate adaptation from receiver-fed-back per-bin SNR, plus ARQ for residual block errors at higher MCS.
- *Mobility.* Everything here assumes the static contact-range channel. For devices in relative motion the channel becomes doubly dispersive, and the delay–Doppler waveforms (OTFS) studied for underwater acoustic channels [18, 19] are the natural candidates — untested here.

8.2 Environment-adaptive scheduling and user guidance

Two of the headroom items above—rate adaptation and the human factors of a contact-range link—motivated decision surfaces that are now implemented in the library as pure, unit-tested logic (the sensing that would feed them with live metrics still requires a sounding burst, i.e. audio, and is the open half).

An MCS ladder driven by SNR and delay spread. The sounder maps a measured median in-band SNR and a -15 dB delay spread to the most aggressive modulation/rate whose two gates both clear, walking 16-QAM $r3/4 \rightarrow$ 16-QAM $r1/2 \rightarrow$ QPSK $r1/2 \rightarrow$ BPSK $r1/2$ and falling to a non-coherent multitone-FSK floor when the delay spread exceeds the cyclic-prefix cap (the reverberant-desk regime). It sizes the CP to cover the measured delay spread plus 25% headroom (clamped to a 32 ms cap, switching NFFT $2048 \rightarrow 4096$ past a 1024-sample CP) and—importantly for a consumer link—*never refuses to connect*: when conditions are poor it still returns the robust floor and merely flags that repositioning would help. A companion per-bin bit-loading rule assigns 0/2/4/6 bits by per-bin SNR thresholds (5/15/23 dB), the mechanism behind the “per-bin adaptive bit loading” headroom item. The thresholds are the operating points observed in the channel characterization (Section 3); the ladder is open-loop today, but it is the decision core a closed feedback loop would call.

A measured caveat on SNR estimation. The ladder consumes a per-bin SNR; supplying an accurate one from a brief sounding is the hard part, and a first over-the-air measurement shows the obvious estimator is unreliable. Estimating per-bin SNR as the variance of the channel across a burst of repeated known symbols is *pessimistic*: the -10 to -25 ppm clock offset between the independently clocked devices rotates the channel from symbol to symbol, and that rotation is counted as noise. At the palm-rest cell this estimator reported a median 10.5 dB and recommended QPSK, while the channel in fact carried 16-QAM rate $3/4$. Removing the per-symbol phase ramp before measuring variance over-corrects the other way (to ~ 30 dB, which would select a 64-QAM loading that does not decode), because a sounding of identical low-PAPR pilots never excites the channel-estimation error and PAPR-driven nonlinearity that random data incurs (Section 6.5). Repeated-pilot statistics, in short, do not predict random-data SINR. We therefore replaced the estimator with a *data-representative* probe—a known 16-QAM frame decoded through the real receiver and read out as EVM—with EVM-to-MCS thresholds calibrated over the air. The corrected sounder is *exact* where the old one under-called: at the palm-rest cell it now recommends 16-QAM rate $3/4$ (the measured ceiling) instead of QPSK, picks 16-QAM rate $1/2$ at a noisier cell, and drops to the non-coherent floor only when the EVM probe (run at the adaptive CP, Section 8.3) confirms

no coherent tier decodes—validated across four geometries, head-to-head against the old estimator. The fix touches only the SNR-*estimation* half; the ladder logic itself, given a correct SINR, is a few lines.

Turning channel metrics into a human instruction. Because the channel is dominated by physical placement—contact coupling, clipping at high volume, a hard reflective surface—small user actions move the operating point more than any coding change. The library therefore exposes a guidance function that maps measured metrics to a single prioritized, actionable hint: clipping (received peak near full scale) → *lower the volume*; ultrasonic band with low phase coherence → *switch to the audible band* (Section 8.5); delay spread beyond twice the CP → *rest the phone on a soft surface*; weak received level → *move the devices closer*; high-frequency roll-off → *aim the phone’s bottom edge at the speaker*. The priority order encodes which defect dominates when several fire at once. This is a small piece of code, but it reframes a finicky physical link as something a non-expert can actually seat: the same measured-channel-first philosophy that drove the modem design, surfaced to the end user.

8.3 Robustness across topologies: a non-coherent floor, microphone diversity, and an adaptive cyclic prefix

The contact-range channel is acutely sensitive to placement: sliding the phone’s overhanging edge ~5 cm across the keyboard flipped the measured -15 dB delay spread from 3 ms to 42 ms, and a naive receiver—a single fixed microphone and a 16 ms cyclic prefix—collapsed from 39 kbps to a hard zero across that move. We found that cliff to be mostly a *receiver* artifact, not a channel limit, and close it with four measured mechanisms; together they turn the cliff into a graceful slope.

A non-coherent floor. When the delay spread exceeds what a cyclic prefix can cover, coherent OFDM cannot decode at any MCS, yet the channel still carries energy. A non-coherent multitone-FSK mode—16-FSK across eight interleaved blocks per symbol, per-tone energy detection, symbols longer than the delay spread (so the previous symbol’s reverberant tail decays before detection), with a 3× majority-vote code—survives exactly this regime, being immune to both phase incoherence and inter-symbol interference. Over the air at a reverberant spot where the coherent codec decoded *zero* of every block, the floor delivered 4/4 frames at 68 bps: degraded, but never zero. (This is the tier the sounder already routed to; previously it had no implementation behind it.)

Microphone diversity, chosen by decode not by loudness. The Pixel’s two microphones see materially different channels—at one position one mic carried a 42 ms delay spread while the other saw only 11 ms. Received *loudness* is not a reliable way to choose between them: at a second position the louder mic failed to decode while the quieter one decoded with zero byte errors, so an RMS-based choice reported a false “dead” link. Selecting instead by decode outcome—try both, keep the one that verifies—is what works.

Two-mic maximal-ratio combining. Better than selecting either microphone is combining both: a per-subcarrier conjugate-weighted sum fills the frequency nulls that sink one mic, because the two mics’ reverberant null patterns are decorrelated (measured: only 1% of bins are nulled on *both* mics, versus 12% on *either*). Over the air at a spot where *both* microphones alone decoded 0 blocks, MRC recovered 8 of 11 QPSK blocks (EVM 1.4 → 0.69)—decoding where neither mic alone could.

An adaptive cyclic prefix. The original 32 ms CP cap bounded overhead, but at the high SNR of these channels, covering the delay spread is a bargain. Sizing the CP to the measured spread (up to 96 ms) rescues reverberant spots into coherent OFDM—a position the old sounder routed to the 68 bps floor instead carries 10.9 kbps of QPSK—and, run the other way, shrinks the CP to 5 ms on a clean 0.4 ms-spread channel, cutting overhead and lifting goodput to 48 kbps, above the

headline figure.

Graceful degradation, demonstrated. With the corrected sounder driving microphone selection, the adaptive CP, and the floor, a closed sound-then-link loop linked across deliberately diverse orientations on the one device pair: ~ 48 kbps face-down at a clean contact spot, ~ 11 kbps at a reverberant overhang, and 5 kbps down to 68 bps at a shadowed on-edge placement below the laptop—never zero, and flagging repositioning when even the floor is marginal. The recurring lesson is that the reverberant “dead” spots were largely artifacts of a single (often the wrong) microphone and a too-short CP. The two microphones supply real spatial diversity, which points directly at the 2×2 -MIMO direction (Section 8), where rich multipath becomes an asset rather than a liability.

Two of these mechanisms have production forms beyond what was measured above. The MRC path ships in the portable C library (Section 4.3). And the floor mode’s $3 \times$ repetition code has been replaced by RS(15,11) over GF(16) with detector-confidence erasures—one code symbol per 16-FSK tone decision—which in the digital reverberant-channel model doubles the floor’s net rate (a 32-byte frame: $68 \rightarrow 138$ bps) while measuring *more* robust at the low-SNR edge (at -3 dB SNR under 40 ms reverb, repetition decoded 5/20 frames, RS 18/20: the erasure flags exploit tone-energy confidence the majority vote discarded).

The over-the-air re-validation of both paths (same device pair, laptop raised on a stand so the desk plane sits in the up-firing speakers’ shadow) delivered the doubled floor—**138 bps, 9/9 frames** at two shadowed placements with probe EVM 1.1–1.6—but only after exposing, and forcing fixes for, three defects invisible to the digital channel model. First, the RS floor’s first over-the-air contact decoded *zero* frames: real reverberation yields low tone-energy confidence on many *correct* decisions, and those false erasures overloaded codewords (2 errors + 1 spurious erasure exceeds the $2e + s \leq 4$ budget) that plain error correction decodes—the decoder now attempts the errors-only decode first and holds erasure assistance in reserve for the genuinely ambiguous low-SNR regime. Second, the floor gained the combining rung the coherent path already had: when neither microphone’s decode verifies alone, the per-microphone tone-energy matrices are summed at equal total frame energy and re-decoded (tone-decision errors proved position-independent across the two microphones). Third, and most consequential: the sounder’s single-microphone EVM probe *cannot see MRC potential*. At the reverberant overhang, both microphones probed undecodable (EVM 0.76/1.33) while the two-microphone MRC probe read 0.25–0.32—so the loop was conceding to the 138 bps floor at a cell where MRC sustains coherent OFDM. With the probe now decoded through the library MRC path as well, and one probe retry at twice the delay-spread-sized CP before conceding (heavy-tailed reverberation carries energy well past the -15 dB spread), the live loop at that cell chose QPSK r1/2 via MRC and carried **11.6 kbps with 0/75 blocks decodable on either single microphone and 75/75 recovered by MRC**—the diversity claim of this section demonstrated end-to-end in the closed adaptive loop, through the shipping library code, at $84 \times$ the floor rate it previously accepted. The methodological lesson mirrors Section 8.2’s: digital-only calibration systematically misses how real acoustic fields fail, and each over-the-air pass on new code has paid for itself.

8.4 The optional security envelope and its rate-dependent cost

A short-range audio channel that carries pairing tokens or small payloads will often want confidentiality and authentication; the project provides an optional X25519 + CTR/HMAC (encrypt-then-MAC) envelope for exactly this. Whether it is worth enabling depends sharply on the chosen MCS, and the bulk PHY makes that tradeoff explicit rather than hard-coding it. Computed from the real codec geometry, the per-frame overhead is a 12-byte nonce plus a 16-byte authentication tag (28 B total) on top of a one-time 80-byte handshake; that costs about **0.15% of goodput at 16-QAM**

Table 7: Single-tone STFT phase-jitter (detrended phase standard deviation), iPhone 17 Pro Max speaker, 8 reps per condition: median [min–max], rad (measured, 2026-06-10). Below ~ 18 kHz the speaker is phase-coherent; above it the phase is scrambled regardless of drive amplitude. The Pixel 7a showed the same pattern.

Tone	Amplitude	Phase-jitter std (rad)	Verdict
8 kHz	0.9	0.007 [0.006–0.013]	coherent
19.5 kHz	0.9	12.8 [7.5–14.0]	incoherent
19.5 kHz	0.4	11.1 [6.4–14.4]	incoherent
22 kHz	0.9	15.9 [9.5–24.2]	incoherent
22 kHz	0.4	18.9 [11.2–26.9]	incoherent

r3/4 (~ 38 kbps, large frames) but as much as **87.5% at the multitone-FSK floor** with the tiny frames that regime implies, and the handshake amortizes in ~ 17 ms at the top tier versus ~ 4.6 s at the measured 138 bps floor. The guidance to a library consumer is therefore conditional: enable it essentially for free on the fast tiers, while on the floor the fixed per-frame cost motivates batching larger frames so the tag amortizes. Dropping confidentiality is an application-specific decision with real security consequences, not a default we endorse; the contribution here is only that the *cost* of the envelope is now a documented function of channel quality—measured from the same sounding the MCS sounder uses—rather than a fixed tax. The envelope is experimental and unaudited.

8.5 Measured result: the inaudible-ultrasonic variant

The original goal (and the obvious product form) is an inaudible link in the 18.5–23.9 kHz band. We subsequently ran the bulk-PHY methodology entirely in that band at 96 kHz sampling [34] (the Mac output chain supports the high sample rate the ultrasonic band requires [32]); the answer is that it is *feasible in one direction but transducer-bound in the other*.

Mac→Pixel works. In the 18.5–23.9 kHz band at 96 kHz (NFFT 2048, CP 768, the chirp preamble moved up into 18.6–23.4 kHz), 16-QAM rate 3/4 delivers **~ 9 kbps** of verified goodput at EVM ~ 0.11 , with every block recovered. The ceiling here is bandwidth, not modulation: a ~ 5.4 kHz coherent band at ~ 22 dB SNR caps near 9–10 kbps, and reaching 20 kbps inaudibly would require either more coherent ultrasonic bandwidth than this Pixel offers or dropping below 18.5 kHz (audible).

Pixel→Mac does not close — and the reason is a clean physical result rather than a tuning failure. The mobile micro-speaker is *phase-incoherent* above ~ 18 kHz, and we now establish this with repeated-run statistics on *two* independent phones rather than single shots. A single-tone STFT phase-jitter measurement on the iPhone 17 Pro Max speaker, 8 repetitions per condition, gives a median detrended phase standard deviation (with [min–max]) of 0.007 rad [0.006–0.013] at 8 kHz, amplitude 0.9 (coherent), versus 12.8 rad [7.5–14.0] at 19.5 kHz amp 0.9, 11.1 rad [6.4–14.4] at 19.5 kHz amp 0.4, 15.9 rad [9.5–24.2] at 22 kHz amp 0.9, and 18.9 rad [11.2–26.9] at 22 kHz amp 0.4 (Table 7). In-band received RMS at the ultrasonic tones is ~ 0.005 , roughly $10\times$ weaker than 8 kHz’s 0.053 — power is present, but the phase is scrambled. The Pixel 7a showed the same pattern (0.055 rad at 8 kHz; ~ 10 rad at 19.5 kHz; ~ 25 –31 rad at 22 kHz). Two independent phones exhibiting the same collapse make this a general consumer-micro-speaker limit above ~ 18 kHz, not a single-device quirk. The speaker radiates ultrasonic *power*, but cannot reproduce it with the phase coherence that OFDM/QAM — or any phase modulation — requires.

This resolves an apparent contradiction in the channel survey: a stationary multitone PSD

reports ~ 27.8 dB “SNR” for Pixel $\rightarrow$ Mac at 18.5–21 kHz, yet coherent OFDM demodulation in the same band yields EVM ~ 1.0 (SINR ~ 0 dB). Power-per-bin and phase coherence are different quantities; PSD-based channel surveys overstate capacity for phase-modulated waveforms on this transducer. A room-tone capture confirmed the band is near-silent (in-band RMS $\sim 5 \times 10^{-8}$), so the limit is the transmit transducer, not ambient noise. (One deployment caveat: our quiet room is not the general case — deliberate near-ultrasonic emitters such as anti-eavesdropping jammers occupy exactly this band [23, 24].)

The conclusion is that a *symmetric* inaudible link is bounded by the mobile speaker, not by the protocol. The fast inaudible direction (Mac $\rightarrow$ Pixel, ~ 9 kbps) is real; the inaudible return path would require a non-coherent modulation (energy-based MFSK or OOK, as ggwave uses) to tolerate the micro-speaker’s phase incoherence, at much lower rates.

8.6 Building on this work

For readers who want to extend the system rather than reproduce it, the open directions below each have scaffolding already in the repository. They are grouped by what they would mature.

Toward a transport. Real-time streaming receiver: move from record-then-decode to a ring-buffer live decode with bounded latency; beyond the buffer plumbing, the substantive change is synchronization—a continuous sliding-window correlator that holds timing lock across stream boundaries rather than the current per-capture matched filter. *A closed adaptive link layer:* wire the sounder, MCS/CP selection, microphone selection, MRC, the floor mode, and block-level ARQ into a session-level MAC, and score it not on peak goodput but on *reliable transfer across changing placements*—a harder and more honest metric than any single number in this paper.

Toward empirical breadth. A device/geometry matrix: a systematic campaign across laptops, phones, surfaces, distances, orientations, ambient noise, and volume settings would turn “works on these devices” into an empirical map of commodity acoustic channels. *An open channel corpus:* publishing raw captures with metadata, decoded block maps, SNR/EVM/delay-spread estimates, and the negative findings would let others evaluate receivers, equalizers, sounders, and learned demodulators without owning the bench. *Motion robustness:* hand motion, pickup, and typing vibration are untested; if coherent OFDM collapses under motion, the candidates are shorter frames, denser pilots, time-domain equalization, or delay–Doppler (OTFS-style) modes.

Toward signal-processing depth. Per-bin bit loading: the receiver already computes per-subcarrier SNR every frame and the demapper accepts non-uniform bits-per-bin, so loading is a transmitter-policy change, not a receiver rewrite—and the interesting version is *effective-SINR* loading calibrated on measured EVM, which predicts which bins survive real transducer phase noise rather than which have raw SNR (Section 6 shows those diverge). *Generalized microphone diversity:* MRC is measured on one phone’s two microphones; laptop microphone arrays, iPhone microphone selection, and per-band combining are unexplored. *2 $\times$ 2 MIMO:* the measured decorrelation between the two microphones (Section 8.3) and between the two laptop speakers (defect 2) is the raw material for channel-matrix estimation, MMSE/ZF stream separation, or SVD precoding; the sounder framework is the natural home for per-speaker pilot separation. *A hybrid coherent/non-coherent modem:* coherent OFDM as the fast path and the RS-coded MFSK floor as the universal fallback, with smooth transitions—a link that is never fast everywhere but nearly always works. *Stronger codes:* we deliberately kept the $K=7$ convolutional code because it has never been the measured bottleneck; the honest trigger for LDPC or polar codes is a measured operating point where it is.

Toward products. An asymmetric inaudible protocol: symmetric ultrasonic QAM is physically blocked by the phone speaker (Section 8.5), but a fast ultrasonic downlink with a low-rate non-coherent return and an authorized audible fallback is not. *A secure pairing layer:* authenticated

encryption, replay protection, and explicit pairing UX on top of the existing envelope, with the overhead measured honestly at every MCS (Section 8.4). *A withheld physical benchmark*: formalize Section 9—fixed repository, hidden placements, device pairs, and payloads, ordered byte verification, goodput delta plus a claims-integrity audit—into a standing evaluation for coding agents.

Wider ultrasonic band at 96 kHz: the hardware qualifies on some pairs (Section 8); the binding constraints found so far are transducer, not algorithmic. In every case the negative-findings log (`docs/NEGATIVE_FINDINGS.md`) records the dead ends already paid for—the cheapest contribution this project can make to a successor is that they not be paid for twice.

9 AI-Agent Development, and the Project as a Capabilities Benchmark

This system was developed almost entirely by AI coding agents operating the laboratory bench through scripted device control (`adb` on Android, `devicectl` on iOS, CoreAudio on the Mac; Section 11), under human direction that was largely product-level: choosing objectives, placing phones on palm rests, reviewing and merging pull requests. Such provenance is usually a disclosure footnote. We promote it to a section because the project turned out to function as an unusually clean *capabilities benchmark* for the agents themselves, and because the repository happens to record which agent built which stage—Claude-era commits carry model-identifying trailers—so the comparison can be reported with the same claims-versus-measurement discipline as the rest of the paper.

9.1 What each era delivered

Table 8 summarizes the development eras. Attribution for the 2026 February–May era is from the operator’s records (sessions with OpenAI GPT-5.5 tooling, which did not stamp commit trailers); attribution for the June and July eras is from Co-Authored-By commit trailers in the repository history, verifiable with `git log -pretty="%ad %(trailers:key=Co-Authored-By)"`.

9.2 The differentiator was verification discipline, not code fluency

Every era produced fluent, sophisticated-looking signal-processing code. The February–May stack implemented an ARQ transport, a chirp-spread-spectrum control channel, an ECDH security envelope, and a 2×2 MIMO handshake—and its documentation celebrated a “100% working” bidirectional link on the strength of three decoded messages in thirteen transmit loops, alongside a “symmetrical 20+ kbps” figure computed from subcarrier counts rather than measured (`walkthrough.md`, preserved in-repo under a correction banner). The measured rate was under 0.3 kbps. Nothing in that gap is specific to acoustic modems; it is the canonical failure mode of agentic engineering—optimistic self-assessment unchecked by external measurement—expressed through a physical channel that happened to keep the receipts.

The June era’s first acts were to characterize the channel (Section 3) and to adopt the conservative goodput definition of Section 6; every subsequent claim was gated on an over-the-air, byte-verified capture, and dead ends were recorded as first-class negative results. Same human operator, same hardware, same repository. We do not think the later models simply “knew more DSP”—the first era’s DSP was largely textbook-correct in isolation. The observable difference was behavioral: whether the agent treated its own throughput claims as hypotheses requiring physical evidence.

Table 8: Development eras by agent. “Claimed” is what the era’s own documentation asserted at the time; “measured” is over-the-air and byte-verified unless marked digital. Commit counts exclude merge commits.

Era	Agent	Built	Claimed	Measured
Feb–May 2026 (68 commits, episodic)	GPT-5.5 (operator records)	Ultrasonic transport stack: gears, ARQ, streams, chirp-spread control channel, 2×2 MIMO handshake, ECDH envelope, room-tone notcher	“Symmetrical 20+ kbps”; “Gear 4 ~12 kbps”; “100% working bidirectional link”	<0.3 kbps (24 B per ~700 ms)
Jun 9–10 (37 commits, 2 days)	Claude Fable 5 (trailers)	Channel characterization; wideband bulk PHY; the four-defect diagnosis; on-device Kotlin verification; iPhone port	—	36.6 / 27.3 kbps (Pixel); 36.6 / 16.9 kbps (iPhone)
Jun 10–12 (33 commits, 3 days)	Claude Opus 4.8 (trailers)	Portable-C port pinned by golden vectors; EVM-probe sounder; adaptive CP; mic selection; MFSK floor; two-mic MRC validation; measurement campaign	—	39.3 kbps library-native; 48 kbps → 68 bps graceful degradation
Jul 1–2 (2 days)	Claude Fable 5 (trailers)	MRC and an RS-coded floor ported into the C library (golden-pinned); library-native adaptive loop; project website	digital validation only, stated as such	(OTA: next row)
Jul 6–8 (3 days)	Claude Fable 5 (trailers)	Over-the-air re-validation of the diversity stack; three defects invisible to digital calibration found and fixed (floor erasure policy, floor stereo combining, MRC-aware sounding + CP escalation)	—	138 bps RS floor (9/9 frames); 11.6 kbps with 75/75 blocks MRC-carried where neither microphone decoded alone

9.3 Why this task is a good agent benchmark

Most coding benchmarks are corpus-derived: the answers exist, so saturation and contamination are permanent concerns. This task has five properties that are hard to get elsewhere at once:

1. **The answers do not pre-exist.** The quantity being optimized—measured goodput between two specific transducer chains in a specific geometry—had no published value until this project measured it; with publication, these numbers become the published *baseline*, not the score. Evaluations stay contamination-resistant because the score is re-measured, never recalled: grade future runs on withheld placements, device pairs, and payloads chosen after submission, so training-data recall of this paper reproduces the starting line, and any improvement still has to be produced over the air.
2. **Physics grades the work.** The score is byte-verified bits over the air. It cannot be inflated by eloquent reports, test-shaped code, or self-assessment—the failure mode the February–May

era exhibited is exactly what the metric catches.

3. **The difficulty ladder is dense and open-topped.** Digital loopback, clean-geometry OTA, adverse placements, diversity combining, MIMO spatial multiplexing—with a Shannon estimate of 155–184 kbps (Section 6) leaving $4\times$ headroom above the best result to date.
4. **It exercises the full agentic loop.** Hypothesis formation, instrument building, physical measurement, revision under ambiguous evidence, and honest reporting—including whether the agent rediscovers dead ends the negative-findings log already paid for.
5. **Sessions are comparable.** The harness (golden vectors, offline self-tests, scripted phone control, one JSONL row per run) makes independent attempts scoreable against each other.

9.4 A benchmark protocol, and the limits of the comparison above

A calibrated version of this benchmark is straightforward to state: fix a repository commit, a hardware pair, a placement set, and a budget (wall-clock sessions or tokens); the agent may do anything the harness allows; the score is the measured, byte-verified goodput delta across the placement set, paired with an *integrity audit*—every throughput claim in the agent’s output matched against the measurement that does or does not back it. The second component matters as much as the first: this project’s history says the dominant failure mode is not inability but overclaiming.

The comparison in Table 8 does not meet that standard, and we present it accordingly: it is an $n=1$ longitudinal case study, not a controlled trial. The eras were sequential, so later agents inherited scaffolding, recorded negative results, and an operator with accumulated prompting skill; budgets were not equalized; the February–May era also pursued a harder inaudibility constraint before the June era deliberately abandoned it. The June era partially controls for inheritance by discarding the first era’s PHY wholesale rather than repairing it. What the record supports is an existence proof, not a ranking: on the same hardware, in the same repository, under the same operator, agent eras separated by months differed by two orders of magnitude in measured-versus-claimed throughput, and the separating variable visible in the artifacts is measurement discipline. The protocol above is what would make future comparisons fair; the repository, harness, and metric are published so that anyone—including model developers—can run them.

10 Summary of Findings

1. A laptop and a phone at contact range can exchange tens of kbps acoustically: **36,571 bps** Mac→Pixel 7a and **27,307 bps** Pixel→Mac, byte-verified over the active message span — $\sim 137\times$ the best recovered open-source baseline we ran on the *same* channel and geometry (ggwave at 267 bps), and $4.5\text{--}9\times$ the original design document’s own projection. The headline enabler was abandoning the inaudibility constraint, which widened the channel from 5 kHz to 16–22 kHz.
2. The measured channel, not the modeled one, dictated every major design parameter: CP length (delay spread is $\sim 10\times$ the modeled value), pilot structure (clock skew, not Doppler, is the always-present phase enemy), speaker selection, and band edges (Pixel→Mac dies above 17 kHz).
3. Four stacked, simulation-invisible defects each individually killed the link; cheap discriminating experiments — including negative results that exonerated nonlinearity and the playback path

— isolated them. Per-symbol pilot-EVM LLR weighting (soft erasures) was the highest-leverage single fix.

4. Verification discipline matters: a deterministic cross-language PRBS lets the receiving phone prove goodput on-device, and a conservative goodput definition (ordered, byte-verified payload over the active message span) keeps the result honest. The $80\times$ gap between an earlier capacity-formula projection and its measured goodput is the cautionary baseline.
5. Consumer audio platforms are an active part of the channel: Android silently zeroes capture based on UI visibility, processed capture paths corrupt QAM, and macOS fades stream tails. Budget for the OS.
6. A measurement can be made reusable without losing its rigor: the full codec ported into portable C, pinned by a committed golden-vector contract under a tiered exact/float tolerance (with an Apple vDSP backend gated by the same vectors), decodes over the air byte-verified through the library itself at 39.3 kbps—*exceeding* the harness headline on the same hardware. On top sit two decision surfaces—an SNR/delay-spread MCS sounder that never refuses to connect, and a channel-metric repositioning guide—that turn the project’s measured-channel-first philosophy into APIs an application can call.
7. The link degrades *gracefully* across topologies, and most “dead” reverberant spots were receiver artifacts, not channel limits. The sounder’s first SNR estimator (repeated-pilot variance) proved unreliable and was replaced by a data-representative EVM probe, exact across geometries; with decode-based microphone selection, an adaptive cyclic prefix, two-mic maximal-ratio combining, and a non-coherent multitone-FSK floor, a closed loop linked everywhere it was tried— 48 kbps at a clean spot, through an 11.6 kbps rescue carried entirely by microphone diversity, down to a 138 bps RS-coded floor, never zero—where a naive single-mic, short-CP receiver collapsed to silence (Section 8.3).
8. The largest remaining gap is empirical breadth, not algorithmic depth. Every result is one contact-range geometry in a quiet room; a distance/orientation/surface sweep and noisier ambient conditions would say more about where this link generalizes than any further modulation or coding work, and is the priority next step.
9. The project doubles as a hard-to-game benchmark for AI coding agents: the optimization target had no pre-existing answer, physics scores the output, and the commit-trailer record shows agent eras on the same hardware separated by two orders of magnitude in measured-versus-claimed throughput—with measurement discipline, not code fluency, as the visible separating variable (Section 9).

11 Reproducibility

Hardware. MacBook Pro M4 (Mac16,5), macOS 26.5 (build 25F71); Google Pixel 7a, Android 16; iPhone 17 Pro Max (iPhone18,2). Signing team Primatech Paper Co LLC (V83C69HYSQ).

Modem configuration. 48 kHz PCM16; NFFT 2048; CP 768; comb pilots on every 8th used bin; 16-QAM; $K=7$ (171,133) convolutional code punctured to rate $3/4$; frame-wide interleave; CRC32 per 256-byte block; chirp preamble plus 2 sync symbols. Volume settings: Mac output 100%, Mac input $\sim 22/100$, phone media at maximum.

Repro commands (research harness). `scratch/hw20k/ota_test.py` (Pixel); `scratch/hw20k/ios_ota_test.py` and `ios_campaign.py` (iPhone); `baselines.py` (gg-wave/minimodem); `ios_phase_coherence.py` (phase jitter). iOS HIL details in `docs/IOS_HIL.md`.

Repro commands (portable-C library). `swift test` runs the codec against the committed golden vectors on the portable KISS-FFT default; `scripts/test-accelerate.sh` re-runs the same vectors through the Apple vDSP/Accelerate backend. `swift run cyrinx-example-adaptive-bulk` exercises the BulkPHY codec round trip, the MCS sounder, and the repositioning guidance. The library-native OTA (Section 6.5) is driven by `scratch/hw20k/ota_clib.py`, which ctypes-loads `libcyrinxbulk` (built from `Sources/CCyrinx/`); `clib.py` is its digital-loopback self-test. The crypto cost figures are in `docs/CRYPTO_TRADEOFF.md`.

Branch commits at time of measurement. `ios-hil-bulk 0f80d8e`, ultrasonic-band 7699a6e, whitepaper d45ccbc.

Development provenance. Agent attribution per era (Section 9) is recorded in Co-Authored-By commit trailers from June 2026 onward:

```
git log --pretty="%ad %(trailers:key=Co-Authored-By)".
```

The February–May era predates the convention and is attributed from the operator’s session records. The first era’s own contemporaneous claims are preserved in `walkthrough.md` under a correction banner. Figure 2 is generated from the reference modem by `scripts/gen-paper-figures.py`.

Artifact paths. Under `scratch/hw20k/data/`: `ios_campaign.jsonl`, `baselines.json`, `channel.json`, and `ios_phase_coherence.json`. The full research writeup, lab notebook, and golden-vector harness are in the repository [35].

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